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Oxygen uptake in the isolated canine brain

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GILBOE, DAVID D., AND A. LORRIS BETZ. *Oxygen uptake in the isolated canine brain.* Am. J. Physiol. 224(3): 588–595. 1973.— Dogs undergoing surgical isolation of the brain were anesthetized with halothane or sodium pentobarbital. Brain O_2 and glucose uptake as well as CO_2 efflux were determined at various rates of glucose delivery after isolation. Oxygen uptake and CO_2 efflux in the halothane group were considered normal because this anesthetic evaporates in the oxygenator, while corresponding values for the sodium pentobarbital group were lower because of the continued presence of anesthetic. Average O_2 uptake in milliliters/100 g per min was 3.39 ± 0.12 SE for halothane and 2.54 ± 0.06 for sodium pentobarbital. Average CO_2 efflux in milliliters/100 g per min was 3.50 ± 0.17 for halothane and 2.52 ± 0.30 for sodium pentobarbital. Cerebral O_2 consumption was constant and unaffected by the rate of glucose uptake. Neither epinephrine nor pentylenetetrazol directly stimulates cerebral O_2 uptake. The initial elevation in cerebrovascular resistance observed in these studies is attributed to high O_2 tensions while the slow increase during perfusion may be due to 5-hydroxytryptamine release from platelets.

anesthetics and cerebral oxygen uptake; canine plasma electrolytes; cerebrovascular resistance; epinephrine, pentylenetetrazol, and sodium pentobarbital and cerebral oxygen uptake; 5-hydroxytryptamine; oxygen glucose index; respiratory quotient

THE BRAIN'S NORMAL O_2 REQUIREMENT has not been well documented due, in part, to the difficulty in establishing and maintaining the proper physiologic conditions for its determination in vivo. The values given in the literature for normal cerebral O_2 utilization range from 3.02 (7) to 5.85 (53) ml/100 g per min. Cerebral O_2 uptake is reduced in the presence of anesthetics (1, 10, 21, 40, 41, 62), brainstem lesions (12, 56), hypothermia (46), and severe anemia (23, 48). Severing the spinal cord may (14) or may not (25) have the same effect. Chemically induced seizures (57), brainstem stimulation (14, 44) and high blood epinephrine levels (33, 57) reportedly have the opposite effect. In the isolated brain, anesthetic effects can be eliminated by using an inhalation anesthetic that readily evaporates from the blood after isolation. Therefore, with the exception of a need to sever the spinal cord, most factors reported to affect cerebral O_2 consumption can be maintained at normal levels.

The major objective of this study was to compare the rate of oxygen consumption by the isolated canine brain under various conditions with corresponding values reported from determinations made in situ. The two anesthetics used in this study, sodium pentobarbital and halothane (Fluothane), were compared with respect to their effects on cerebral O_2 uptake after brain isolation. In some experiments the normal rate of glucose delivery to the brain was altered in order to observe the effect of varied rates of brain glucose uptake on cerebral O_2 consumption. Studies performed in the presence of pentylenetetrazol (Metrazol) and epinephrine were used to determine whether either of these substances stimulates brain O_2 uptake.

MATERIALS AND METHODS

These studies were conducted on 53 brains that were isolated from adult mongrel dogs weighing 15–20 kg. At the time of anesthesia, the subject received phenoxybenzamine HCl (Dibenzyline), 1.0 mg/kg body wt, to prevent surgical shock. The surgical procedure used for isolation (15, 18) left the brain within the cranium. Arterial blood was supplied through the internal carotid arteries and the anastomotic branch of the internal maxillary arteries. Venous blood was collected through a threaded Luer connector that was cemented over a small hole drilled into the bone covering the confluence of sinuses.

The isolated brains were perfused with blood from compatible donors who received exactly the same medication and anesthetic as the corresponding subject. In addition, each donor was given sufficient 6% dextran to reduce the hematocrit to approximately 25%. After the dog was anticoagulated with heparin, blood was pumped from the femoral artery through a 4 x 8 cm Dacron-wool filter (provided by E. I. Dupont de Nemours & Co., Wilmington, Del.) (61) and returned to the contralateral femoral vein. This conditioning process continues for 3–4 hr to reequilibrate the diluted plasma with the low-molecular-weight compounds in the tissues and remove platelet aggregates, fibrin particles, and other debris.

The micro disc oxygenator system was primed with 600 ml of fresh conditioned blood. Arterial pH was maintained at 7.40 and P_{CO_2} at 40 mm Hg by addition of $NaHCO_3$ or adjustment of the CO_2 flow rate to the oxygenator. Arterial P_{O_2} was approximately 500 mm Hg as a result of oxygenation with 97% O_2. To provide an appropriate range of arterial glucose delivery rates, the blood glucose concentration was increased in some experiments by addition of 5% glucose in 0.9% saline. Blood removed from the system during sampling was replaced with fresh donor blood. The arterial blood was propelled by a variable-speed roller pump from the oxygenator, through a Dacron-wool filter, a heat exchanger, and a bubble trap to the brain. The pump speed was set to produce an initial mean arterial

pressure between 80 and 100 mm Hg and the hot water to the heat exchanger was adjusted to a temperature sufficient to maintain the blood supplying the brain at 37 C. Experience has shown (16) that the brain will remain at 37 C under these conditions. The venous blood returned from the brain to the oxygenator by gravity.

Carotid artery pressure was monitored from a transducer connected proximal to the bubble trap. Arterial input and venous output temperatures were determined with flexible thermistor probes. The pumps supplying arterial blood to the brain were calibrated after each study. Bipolar recordings of right and left frontal, parietal, and occipital electrical activity were made intermittently on an electroencephalograph to monitor the viability of the preparation.

Since this study was conducted in conjunction with several other studies, there was some variation in the sampling routine and the total length of the sampling period. To minimize these variations and simplify statistical analysis of the data, only the first four simultaneous arterial and venous P_{CO_2}, P_{O_2}, pH, hemoglobin, and hematocrit determinations from each brain are included in the calculation of values appearing in Tables 1 and 2. In most cases sampling commenced 45 min after the start of brain perfusion and continued at 15 min intervals for 45 min.

One group of 25 animals received 60 mg of morphine sulfate 30 min prior to being anesthetized with 25 mg/kg body wt of the long-acting barbiturate, sodium pentobarbital (Nembutal). This anesthetic was supplemented as required throughout the procedure. Since sodium pentobarbital is detoxicated mainly by the liver, brains isolated from animals receiving this drug would be expected to have the same oxygen requirement as brains in situ under the influence of sodium pentobarbital. Glucose and oxygen uptake were determined simultaneously in all 25 of these isolated brains.

A second group of 25 animals was induced and maintained throughout the isolation procedure with halothane. Halothane was chosen because it is a volatile, nonexplosive anesthetic which rapidly evaporates from the perfusion blood as it passes through the oxygenator. Glucose uptake was determined in 18 of these preparations, while arterial and venous Na^+, K^+, Ca^{++}, Mg^{++}, and Cl^- were determined in nine of the brains in this group. The brain tissue lactic acid concentration in four brains was assayed enzymatically (26) from small plugs of tissue frozen in situ.

After collection of the control data, the following studies were made on eight of the brains isolated with halothane anesthesia:

A) In each of five isolated brains, four separate seizures were induced at 15-min intervals with 0.25-ml intracarotid bolus injections containing 25 mg of pentylenetetrazol. Oxygen uptake and CO_2 efflux were determined during the peak of the electroencephalographic (EEG) seizure activity. None of the venous blood was permitted to return to the oxygenator from the time of the injection until 60 sec after all EEG indication of seizures had stopped, in order to give the brain an opportunity to recover between studies.

B) A second group of three isolated brains received a continuous intracarotid infusion of epinephrine at the rate of 400 ng/min, beginning immediately after the four control samples were taken. Four separate O_2 uptake and CO_2 efflux determinations were made at 15-min intervals during the infusions. The rate of epinephrine infusion was chosen to simulate the epinephrine level found in dogs during respiratory acidosis (20).

Three brains were isolated with halothane anesthesia, but the sampling routine did not permit their inclusion in the other halothane group. In this study, the blood flow rate was increased or decreased in a random manner to determine whether cerebral O_2 uptake varied with the rate of O_2 delivery. Arterial glucose was maintained at about 110 mg/100 ml by the continuous infusion of glucose into the oxygenator. Twelve separate O_2 uptake determinations were made for each brain at 15-min intervals on arterial and venous samples obtained 60 sec after the change in blood flow rate.

Blood pH, P_{CO_2}, and P_{O_2} were determined with appropriately standardized electrodes. The blood glucose concentration was established from triplicate analyses of blood samples with a glucose oxidase method (Fermco test, Fermco Division of G. D. Searle & Co., Skokie, Ill.). Plasma chloride was measured by the method of Schales and Schales (55). Calcium, magnesium, potassium, and sodium were determined on diluted plasma by atomic-absorption spectrophotometry. Blood hemoglobin was converted to cyanmethemoglobin for spectrophotometric assay and microhematocrits were estimated on the original samples. The brains were removed from the skull and weighed at the termination of each study.

All experimental values were recorded on data-processing cards to facilitate their reduction, statistical analysis, and retrieval. The O_2 content in milliliters per 100 ml of blood was calculated from the corresponding P_{O_2}, pH, and hemoglobin values by a subroutine (28) that simultaneously computed the amount of O_2 bound to hemoglobin, from the O_2 dissociation curve for canine hemoglobin (54), and the O_2 dissolved in blood water. The CO_2 content in milliliters per 100 ml of blood was calculated from the individual P_{CO_2}, P_{O_2}, pH, and hemoglobin values by a program described by Kelman (29), which is based on the Singer-Hastings nomogram (59). The Singer-Hastings nomogram assumes that the "plasma protein-bound base" levels are normal (64); however, it was necessary to recalculate the plasma protein-bound base to correct for dilution of the plasma protein. Carbon dioxide efflux and O_2 uptake were calculated by multiplying their respective A-V differences by the rate of blood flow: the values are expressed in milliliters per 100 g of brain tissue per minute. The statistical significance of differences in the rates of either O_2 uptake or CO_2 efflux was tested by calculating the variance ratios.

Glucose uptake was determined by multiplying the A-V difference by the rate of blood flow and expressed in milligrams per 100 g of brain tissue per minute.

The data from paired O_2 and glucose uptake observations were divided according to the anesthetic used during isolation, and straight lines were fitted to O_2 uptake versus glucose uptake by the method of least squares. Regression analysis was used to determine whether the slope of either line differed significantly from zero. A joint confidence

region was established to compare the slopes and intercepts of the two lines (17).

The average cerebrovascular resistance (CVR = mean perfusion pressure (mm Hg) × brain weight (g)/cerebral blood flow (ml/min) × 100) was calculated for each group at 10-min intervals starting at the time of brain isolation.

RESULTS

The EEG tracings shown in Fig. 1 are typical of those seen after the first 10 min of brain perfusion. When the postisolation EEGs in the group anesthetized with halothane are compared with patterns recorded from normal unanesthetized dogs, one observes faster frequencies from the anterior electrodes and a slight overall elevation in amplitude. Such tracings are typical of light sleep and may be the result of deafferentation (3). The background frequency of the EEGs obtained after sodium pentobarbital anesthesia is slow, with superimposed fast activity, a pattern commonly observed in dogs anesthetized with barbiturates.

Table 1 lists mean values for arterial pH, arterial P_{CO_2}, arterial and venous P_{O_2}, brain weight, blood flow rate, perfusion pressure, and hematocrit for both anesthetic groups in this study. Also included are plasma electrolyte levels in the blood that was used to perfuse nine of the halothane brains.

Neither the arterial O_2 tension nor the hematocrits fall within the normal in vivo range. The arterial O_2 tension was elevated as the result of equilibrating the blood with a gas mixture containing approximately 97% O_2. Such high O_2 tensions are routinely used during cardiopulmonary bypass operations in human subjects without obvious ill effects. The large difference between the average arterial P_{O_2} values in Table 1 is related to conditions within the oxygenator (disc speed and blood depth); however, the arterial O_2 content of the sodium pentobarbital group exceeded that of the halothane group by only 3%. Hematocrit reduction to approximately half its normal value was necessary to improve the flow characteristics of blood used to perfuse the isolated brain (38). Nevertheless, the oxygen-carrying capacity of the perfusion blood is sufficient to meet the brain's O_2 requirements as long as the rate of blood flow remains above 30 ml/100 g of brain tissue per minute.

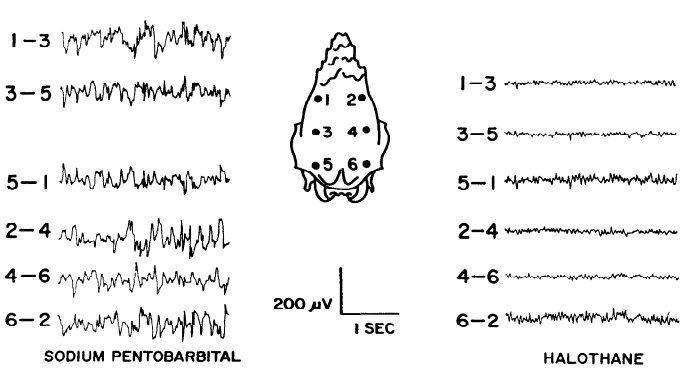


FIG. 1. Electroencephalographic (EEG) tracings typical of those obtained in this study 60 min after brain isolation.

TABLE 1. *Conditions during perfusion of isolated brain*

	Halothane	Sodium Pentobarbital
Arterial pH	7.358 ± 0.003 (100)	7.383 ± 0.007 (100)
Arterial P_{CO_2}, mm Hg	36.0 ± 0.4 (100)	40.0 ± 0.6 (100)
Arterial P_{O_2}, mm Hg	463 ± 8 (100)	587 ± 14 (99)
Venous P_{O_2}, mm Hg	34.6 ± 1.1 (100)	39.7 ± 1.1 (98)
Brain weight, g	85.0 ± 1.1 (25)	87.4 ± 1.5 (25)
Blood flow rate, ml/100 g per min	47.3 ± 0.7 (100)	39.2 ± 1.0 (100)
Perfusion pressure, mm Hg	121 ± 2 (419)	107 ± 1 (469)
Hematocrit, %	22.2 ± 0.4 (100)	23.0 ± 0.6 (100)
Na^+, mEq/liter	146 ± 2 (30) [147 ± 1 (147)]	
K^+, mEq/liter	4.19 ± 0.08 (31) [4.31 ± 0.05 (146)]	
Mg^{++}, mEq/liter	1.75 ± 0.12 (31) [1.35 ± 0.05 (148)]	
Ca^{++}, mEq/liter	4.88 ± 0.03 (31) [4.74 ± 0.02 (148)]	
Cl^-, mEq/liter	118 ± 2 (30) [117 ± 1 (167)]	

Values are means ± SE with the number of determinations shown in parentheses. With the exception of the perfusion pressure and the electrolyte values in brackets, the data were obtained at the moment O_2 uptake and CO_2 efflux were determined. The perfusion pressure is the average of measurements made at 10-min intervals during the first 150 min of perfusion. The electrolyte values in brackets were obtained during experiments conducted under similar conditions, but not included in this study.

Plasma electrolyte concentrations were determined in the blood used to perfuse nine brains in the halothane group (Table 1). The values in brackets are similar and were obtained from 40 studies conducted under identical conditions, but not included in this report. It is assumed that plasma electrolyte levels in the sodium pentobarbital group were comparable. Calcium, chloride, and magnesium are within the range of values reported as normal for the dog while sodium and potassium are slightly below normal (2). Thus, the conditioning procedure (17) probably returns the levels of most low-molecular-weight compounds to near normal after hemodilution. Both arterial and venous samples were taken in all cases, and there was no detectable net movement of Ca^{++}, Mg^{++}, K^+, or Na^+ into or out of the brain during these studies. Significant difference between arterial and venous plasma chloride levels (not shown in Table 1) occurred and was attributed to the chloride shift.

The average values for O_2 uptake, CO_2 efflux, and the respiratory quotient (RQ) are presented in Table 2. Brains from animals anesthetized with halothane had a significantly higher average rate of O_2 consumption and CO_2

TABLE 2. *Oxygen uptake, carbon dioxide efflux and respiratory quotients of isolated canine brain*

	Halothane	*P*	Sodium Pentobarbital
O_2 uptake, ml/100 g per min	3.39 ± 0.12 (100)	<0.001	2.54 ± 0.06 (98)
CO_2 efflux, ml/100 g per min	3.50 ± 0.17 (100)	<0.005	2.52 ± 0.30 (100)
RQ	1.03 ± 0.06 (100)	NS	0.99 ± 0.13 (98)

Values are means ± SE, with the number of determinations shown in parentheses. NS means no significant difference exists between the 2 values that are compared ($P > 0.05$).

efflux than did brains from animals receiving sodium pentobarbital.

It is difficult to determine accurately the volume of CO_2 outflow when electrodes are used to measure arterial and venous P_{CO_2} because the arteriovenous differences in P_{CO_2} are relatively small and the electrodes lack the necessary precision. This causes the CO_2 efflux values to vary more than the O_2 uptake values.

A value of 1.00 is well within the range of the standard error of the RQ in both groups; thus there is no evidence from this study to suggest that the normal RQ for either group is different from 1.00.

Values for O_2 uptake versus glucose uptake were plotted separately for each anesthetic treatment group (Fig. 2). The straight lines fitted to each group of data by the least-squares method appear to be nearly horizontal, an observation confirmed by the regression analysis which indicates that neither slope differs significantly from zero. This means that the rate of O_2 uptake is unrelated to the rate of glucose uptake. Only the intercepts of these lines can be shown to differ significantly from one another by the joint confidence test, supporting the observation that sodium pentobarbital depresses oxygen consumption by the isolated brain (Table 2).

The lactate concentration averaged 1.96 ± 0.29 SE μmoles per g of tissue in the four brains assayed.

There appears to be a direct linear relationship between O_2 uptake and cerebral blood flow rate over the range of values plotted in Fig. 3. The straight line fitted to the data by the method of least squares has a correlation coefficient of 0.91, which strongly suggests that cerebral O_2 uptake increases as the rate of blood flow increases.

Table 3 contains the average blood flow rate, O_2 uptake, and CO_2 efflux values obtained from brains before and during treatment with pentylenetetrazol and epinephrine. The average rate of O_2 uptake for the controls in these studies is slightly above the average for the entire halothane group (Table 2), which is most likely due to the higher average flow rates. Neither drug significantly altered the rate of O_2 uptake or CO_2 efflux.

Plots of average CVR values for the first 160 min of brain perfusion are shown in Fig. 4. The total change in CVR during this period is nearly identical in both groups; however, in the halothane group CVR increases only slightly after 30 min of perfusion while in the sodium pentobarbital group CVR continues to increase at a more or less uniform rate.

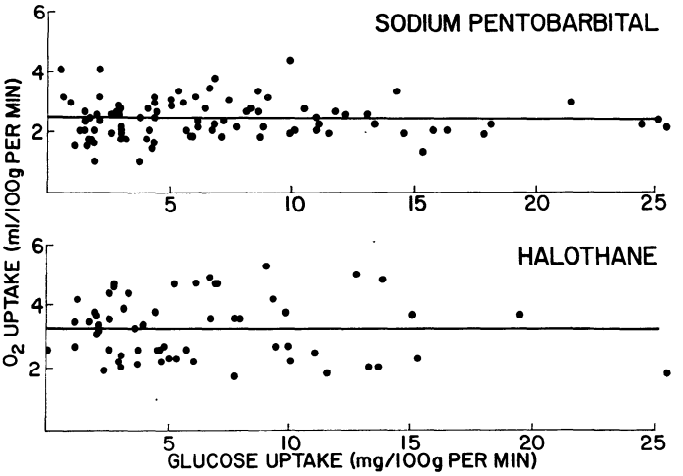


FIG. 2. A plot of O_2 uptake vs. glucose uptake for brains isolated with sodium pentobarbital and halothane anesthesia. Slope and intercept for sodium pentobarbital are 0.0029 and 2.59, respectively, while for halothane they are 0.0056 and 3.37, respectively. Neither slope differs significantly from the other or from 0; however, the intercepts do differ significantly.

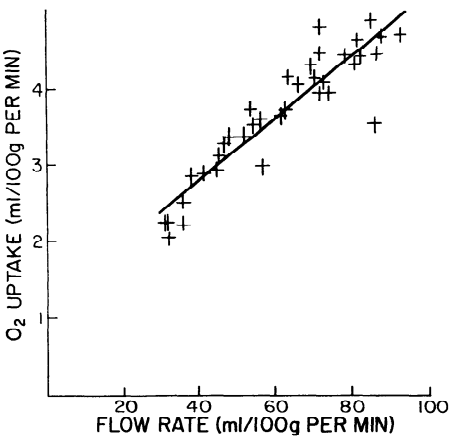


FIG. 3. A plot of O_2 uptake vs. blood flow rate obtained from studies in which cerebral blood flow rate was changed in a random manner.

DISCUSSION

The average of 159 normal cerebral O_2 uptake values reported by various authors (30, 31, 32, 35, 36, 37) for the unanesthetized human is 3.35 ± 0.05 ml/100 g per min. The O_2 uptake value for canine brains isolated with halothane anesthesia (Table 2) is similar because halothane is believed to evaporate from the oxygenator. Few, if any, studies conducted with intact experimental animals are comparable because an anesthetic is usually required to control the animal while the measurements are being made and most general anesthetics cause some reduction in the rate of cerebral O_2 utilization.

In the presence of sodium pentobarbital, O_2 uptake by in situ monkey brain (2.83 ± 0.13 ml/100 g per min, $n = 192$ (44, 57)), isolated monkey brain (<2.25 ml/100 g per min, $n = 3$ (27)), and isolated cat brain (2.24 ± 0.05 ml/100 g per min, $n = 4$ (65)) is depressed. Oxygen uptake of the

TABLE 3. *Oxygen uptake and carbon dioxide efflux before and during pentylenetetrazol and epinephrine administration*

	Flow Rate, ml/100 g per min	Control O_2 Uptake, ml/100 g per min	*P*	Experimental O_2 Uptake, ml/100 g per min	Control CO_2 Efflux, (ml/100 g per min)	*P*	Experimental CO_2 Efflux, ml/100 g per min
Pentylenetetrazol	52.5 ± 2.4 (5)	4.08 ± 0.20 (19)	NS	3.78 ± 0.16 (20)	3.55 ± 0.41 (19)	NS	4.36 ± 0.49 (20)
Epinephrine	72.9 ± 8.4 (3)	3.66 ± 0.18 (8)	NS	3.90 ± 0.14 (12)	5.64 ± 0.97 (8)	NS	4.66 ± 0.71 (12)

Values are means ± SE, with the number of determinations shown in parentheses. NS means no significant difference exists between the 2 values that are compared ($P > 0.05$).

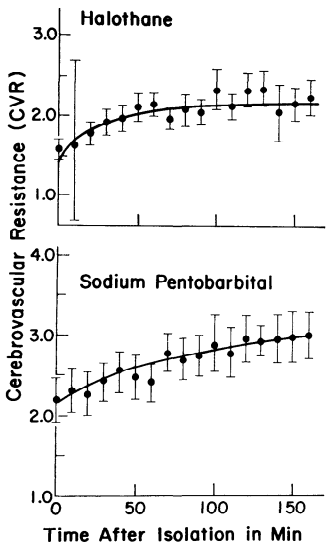


FIG. 4. Mean cerebrovascular resistance (CVR) values bracketed by their SEs are plotted at 10-min intervals starting at time of isolation. A smooth line was drawn on graphs to aid in visualizing trend of CVR increase in each group.

sodium pentobarbital group was lower than the O_2 uptake of the halothane group in our studies (Table 2). This disparity in the O_2 uptake of the two groups is too large to be accounted for on the basis of differences in flow rate (Fig. 3). Morphine also depresses cerebral O_2 uptake in the human (45). Unlike sodium pentobarbital, morphine was not supplemented in the pentobarbital group and its concentration in the brain steadily declined during the isolation procedure. Thus, the depressed O_2 uptake in those brains isolated under barbiturate anesthesia is primarily due to the continued presence of sodium pentobarbital. One study reports a rate of cerebral O_2 consumption (3.37 ± 0.58 ml/100 g per min, $n = 7$) comparable to the halothane group in dogs anesthetized with sodium pentobarbital (12). Although the depth of barbiturate anesthesia influences cerebral O_2 consumption (10, 57), these seemingly high values for anesthetized dog brain may be the result of a failure to correct for the O_2 consumption by extracranial tissues.

When venous blood is taken from the sagittal sinus, estimates of brain oxygen utilization in both unanesthetized (5.59 ± 0.26 ml/100 g per min, $n = 12$ (63)) and anesthetized (4.85 ± 0.31 ml/100 g per min, $n = 165$ (1, 19, 24, 46, 48, 49, 62, 63)) dog brains exceed the value determined for the halothane group (Table 2). The confluence of sinuses and the jugular bulb drain mixed venous blood from the whole brain; however, the sagittal sinus receives its blood primarily from the cortex (19, 62) which is metabolically more active than the white matter. Although some indirect methods of determining the rate of cerebral blood flow tend to give high estimates which would make cerebral O_2 uptake appear high (53), most of the authors cited made direct determinations of the rate of blood flow through the sagittal sinus. The evidence suggests that values calculated from arterial-sagittal sinus O_2 differences are more representative of the O_2 requirement of the cerebral cortex. The discrepancy in mean O_2 uptake values is probably not due to differences in the average age of the experimental animals (47) since adult dogs were used in all studies. Recent work (25) indicates that severing the spinal cord in the dog does not influence cerebral O_2 consumption; therefore, our values for the halothane group probably represent normal O_2 uptake by the whole, waking canine brain.

There are two probable sources of CO_2 in brain tissue: one is metabolism of carbohydrates, lipids, and amino acids, and the other is buffering of metabolically formed acids such as lactic acid by intracellular bicarbonate (34). Intracellular buffering is potentially a more important source of CO_2 from the isolated perfused brain than it is from the brain in situ because many vascular and metabolic problems may arise after isolation that could cause brain lactic acid to accumulate. It is impossible to ascertain the relative importance of either source without some knowledge of changes that are taking place in tissue lactic acid or bicarbonate content when CO_2 efflux is being determined.

Carbon dioxide is generated by all groups of brains at very nearly the same rate that O_2 is consumed. The RQ for the halothane group does not differ significantly from unity (Table 2). Tissue metabolite determinations for a few halothane-group brains indicate that normal levels of lactic acid are present (8), implying that CO_2 must arise primarily from metabolic reactions involving carbohydrates. If the halothane group RQ were significantly higher than 1.00, it would suggest that net synthesis of lipid was taking place. Respiratory quotients slightly greater than unity have been reported for brain tissue (32).

It is apparent from the RQ of 0.99 that the CO_2 efflux values for the sodium pentobarbital group are very nearly identical to their O_2 uptake values. We have shown (4) that significant quantities of amino acids are catabolized by isolated brains in the presence of sodium pentobarbital so that the true RQ is probably below 0.99, but within the range of the standard error of our calculated value. No tissue lactic acid determinations were made in this group, and one cannot make any inference as to the possible contribution of intracellular buffering to CO_2 efflux; however, the rate of lactic acid formation appears to be reduced in the presence of anesthetics (8). It would appear that the method used to determine CO_2 efflux in this study tends to overestimate the rate of CO_2 formation and does not have

the precision necessary to detect small differences in the RQ.

The oxygen glucose index (OGI) has been used to evaluate the state of cerebral metabolism (7) under various conditions. If it is assumed that the O_2 taken up by the brain is utilized exclusively to oxidize the entering glucose to CO_2 and H_2O, then OGI is an expression that denotes the percent of glucose that is completely oxidized. Although most evidence suggests that glucose is not immediately oxidized, the OGI normally gives a reasonable estimate of the overall oxidation rate of glucose and compounds derived from glucose. Values in the range of 85–90% are considered as evidence of a normal pattern of cerebral glucose metabolism, while values below 85% suggest an increasing dependence on glycolysis. If the OGI is to be useful for relating cerebral glucose metabolism during various treatments to normal metabolism, it should remain fairly constant when differing rates of glucose delivery to the brain cause the rate of glucose uptake to vary (17). In other words, cerebral O_2 uptake should change in direct proportion to the change in glucose uptake. Actually, brain O_2 uptake remains constant over the range of glucose uptake values observed in this study (Fig. 2), so there is no demonstrable relationship between the rates of cerebral O_2 and glucose uptake. Thus, one must question the metabolic significance of changes in OGI that are the result of an increase in cerebral glucose uptake induced by a greater rate of glucose delivery to the brain (7).

This study presents a limited number of O_2 determinations obtained at the time of pentylenetetrazol seizures, but the results in Table 3 give no evidence to suggest that the drug had any significant effect on cerebral O_2 consumption at the time the measurements were made. During pentylenetetrazol seizures in sodium thiopental-nitrous oxide anesthetized dogs with high cervical section of the spinal cord, Plum et al. (52) observed a transitory decrease in cerebral venous P_{O_2} accompanied by an increase in cerebral blood flow. Several other investigators (9, 42, 43) have also observed increases in the rate of cerebral blood flow during seizures. Schmidt et al. (57) administered pentylenetetrazol and noted an increase in the average rate of O_2 uptake in the anesthetized monkey brain that was accompanied by a nearly proportional increase in the rate of cerebral blood flow. They also reported that there was a direct relationship between the rate of cerebral O_2 uptake and the rate of cerebral blood flow in the absence of a convulsant. We have confirmed this finding (Fig. 3). The reason for this relationship is unclear, but it may be due to vascular distention at the higher rates of flow resulting in a larger surface for O_2 diffusion.

The rate of blood flow to the brain could be increased either by an increase in systemic blood pressure or by dilation of the cerebral vasculature. Pentylenetetrazol seizures in both dog and monkey caused a large increase in arterial blood pressure (52), while we have noted a significant fall in CVR ($12 \pm 2\%$, $P < 0.01$) in the present study. Plum et al. (52) and Schmidt et al. (57) reported CO_2 efflux to be higher than O_2 uptake during pentylenetetrazol seizures. Although our findings were similar (Table 3), CO_2 efflux did not differ significantly from the control values because of a lack of precision in the method used for its determination. A higher rate of CO_2 efflux could be the result of increased rates of either oxidative metabolism or glycolysis since tissue bicarbonate would be expected to buffer some of the free lactic acid in the brain if glycolysis was stimulated. Stone and co-workers (60) observed a marked increase in brain lactic acid content during pentylenetetrazol seizures, which suggests that it is glycolysis and not oxidative metabolism which is stimulated. Thus, the increase in cerebral O_2 uptake observed during pentylenetetrazol seizures in the intact animal is probably more the result of a transitory increase in the rate of cerebral blood flow than an increase in the brain's O_2 requirement.

When epinephrine was infused into three isolated brains at a rate too low to cause pressor effects on the cerebral vasculature (39), no significant increase in the rate of cerebral O_2 consumption was observed (Table 3). Neither intravenous infusions nor intramuscular injections of epinephrine are reported to cause a significant change in the cerebral O_2 consumption rate of human subjects (11, 58). Schmidt et al. (57) were unable to stimulate cerebral O_2 uptake with epinephrine in those monkey brains that were consuming O_2 at a normal rate; however, they did observe a small but statistically insignificant increase in the rate of cerebral O_2 uptake when the consumption rate was depressed prior to epinephrine administration. In contrast to our findings, King and co-workers (33) reported a statistically significant increase in cerebral O_2 consumption when larger doses of epinephrine were administered to human subjects. However, the increase in cerebral O_2 uptake was directly proportional to the increase in the rate of cerebral blood flow. Again, increased O_2 flux appears to be the result of an increased rate of cerebral blood flow. There is no evidence at present to indicate that endogenous epinephrine directly stimulates cerebral O_2 consumption in vivo. It is doubtful that phenoxybenzamine HCl had a significant effect on these experimental results since it was administered preoperatively at 10% of the rate required to block the α-receptors in the cerebral vasculature (39).

The CVR tends to increase during brain perfusion, but the cause for this increase has not been established. There is no evidence that our filtering system removes microbubbles (51), but it does remove platelet aggregates and other debris (61). If the microvasculature was being blocked to any significant degree by microbubbles, one would expect the rate of cerebral O_2 consumption to decrease (5) and the EEG to deteriorate as the perfusion continues. However, both cerebral O_2 consumption[1] and EEG remained stable.

Halothane is known to act as a vasodilator (6). The time course of the CVR increase is similar to the pattern that one would predict for brain halothane washout; thus, the plot of CVR versus time could be viewed as a dose-response curve, with the lower doses being related in some manner to the longer perfusion times. This explanation, however,

[1] A straight line was fitted by linear regression to all O_2 uptake data collected during the first 210 min after isolation. The slopes and intercepts were -0.00026 ml/100 g per min^2 and 3.70 mg/100 g per min ($n = 163$), respectively, for the halothane group, and -0.0024 ml/100 g per min^2 and 2.58 mg/100 g per min ($n = 245$), respectively, for the sodium pentobarbital group. Neither slope was significantly different from 0, indicating that O_2 uptake remains constant over this time interval.

does not account for the data obtained with sodium pentobarbital, a drug which lacks profound vasodilatory properties and which cannot be eliminated from an extracorporeal brain perfusion system.

The isolated brains were perfused at a constant rate with blood having an unphysiologically high P_{O_2}. It is well established that cerebral vasoconstriction occurs with a high P_{O_2} in the presence of either a low or normal (22, 23, 32, 35) P_{CO_2}. The increases in the halothane and sodium pentobarbital CVRs in the present work are similar in magnitude to those reported for humans breathing 85–100% O_2 (32), but are greater than CVR changes observed by others under similar conditions (22, 23, 35). Since the O_2 carrying capacity of blood is adequate at atmospheric O_2 tensions, there is little need to equilibrate with 97% O_2 when perfusing the isolated brain. Recent studies of 60 isolated dog brains (unpublished results) indicate that there is a statistically significant initial reduction in CVR of 35% when compressed air is substituted for pure O_2 in the oxygenator. Nevertheless, the CVR slowly increases during perfusion.

Historically (50), increases in arterial pressure during extracorporeal perfusion have been overcome by using the lung to destroy 5-hydroxytryptamine (5-HT) released from the platelets (13). We have found (unpublished results) that arterial concentrations of 5-HT as low as 10 μg/liter cause a significant constriction of the cerebral vasculature. Platelets adhere to the filter material, but they are not removed from the system; thus they may be continually releasing 5-HT into the perfusion blood. Although the evidence is tentative, 5-HT release from the platelets provides a tenable explanation for the slow, steady increase in CVR seen in Fig. 4. The higher CVR and lower blood flow rate with the sodium pentobarbital group could be due to higher initial levels of 5-HT in the perfusion blood since there tended to be a longer time interval between priming of the system and the start of perfusion with this group. The increasing CVR does not appear to be a serious problem in our studies, but an effort is underway to eliminate factors which might cause cerebral vasoconstriction during perfusion.

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